What is the Update? Roof Zones were revised for Hip and Gable Roofs

As wind blows over roof surfaces, it creates suction, or uplift, on the roof assemblies. The amount of uplift varies by building height, location (associated wind speed), and other factors unique to each building. The uplift created on the building is not uniformly distributed, and will vary depending on factors such as roof shape. Wind uplift is highest at the corners, then perimeters, and is the least in the field, or center of the roof; these varying wind uplift locations are called roof zones.

Hip Roofs are where all four sides of the roof slopes down to connect to the exterior walls at the eaves. Gable roofs have two slanted sides that form a ridge that connect to the vertical walls that extend to the ridge. Chapter 30 of both ASCE 7-16 and 7-22 include roof zone diagrams and graphics that can be used to determine the External Pressure Coefficients (GC_p).

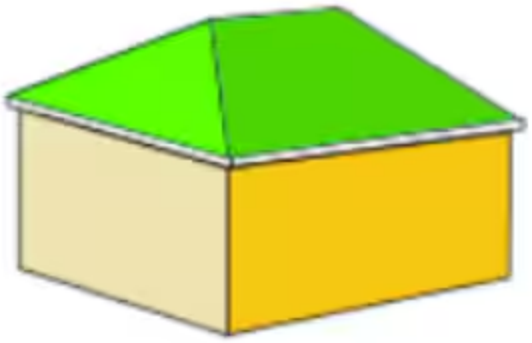


Image 1: Hip roof

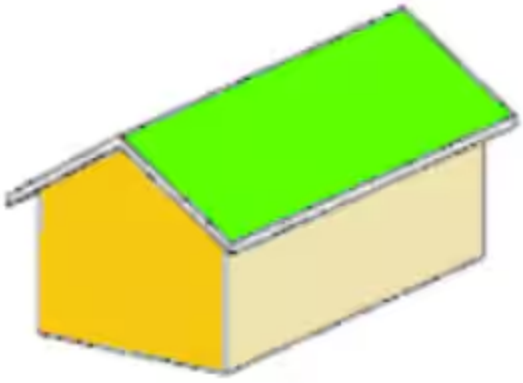
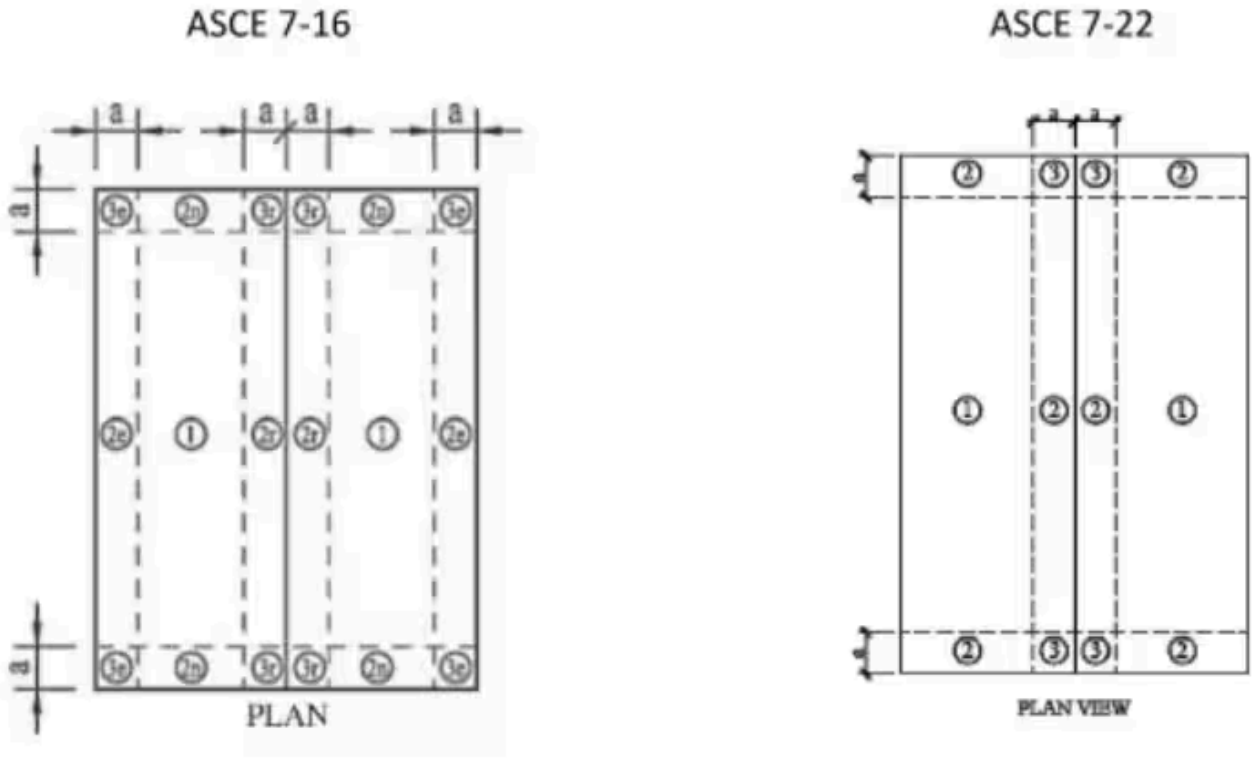


Image 2: Gable roof

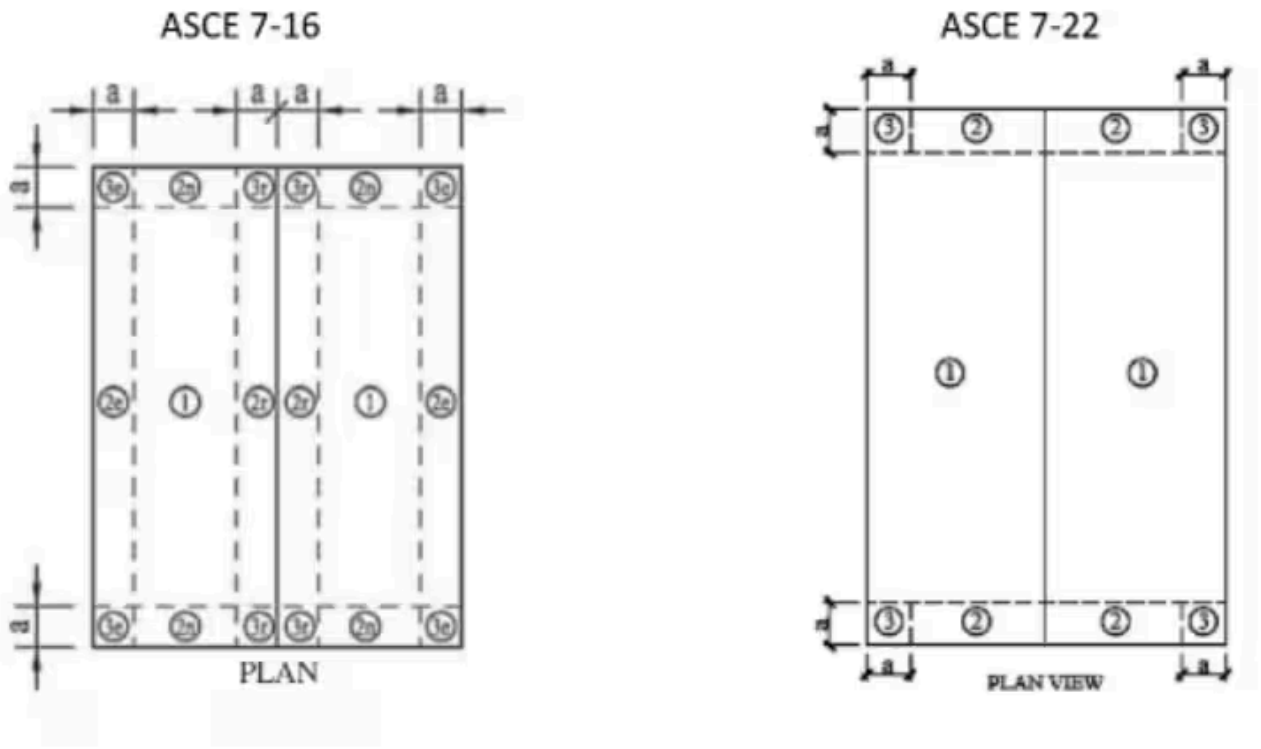
Images 1 and 2: Hip roof and gable roof, images courtesy of roofingcalc.com

In ASCE 7-22, both the roof zone diagrams and the graphics to determine GC_p are updated. The roof zones are simplified to have three zones, Zone 1, Zone 2, and Zone 3 (in lieu of Zone 1, Zone 2r, Zone 2e, and Zone 3), and the accompanying zone layout was modified to include the Zone changes. The External Pressure Coefficients are determined using graphics, and those also have been updated and simplified.

Components and Cladding, h ≤ 60 ft, Gable Roofs Roof Slopes 7°≤Θ≤ 20° and 20°≤Θ≤ 27°

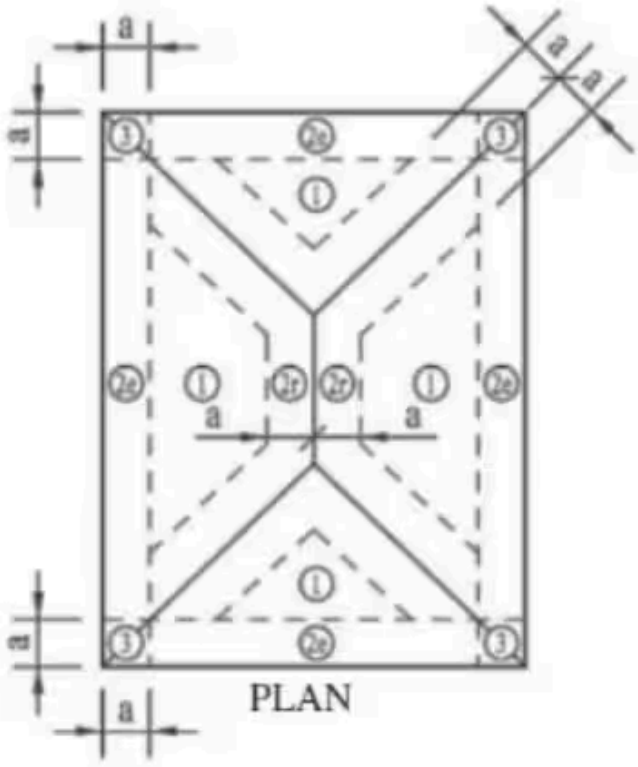


Components and Cladding, h ≤ 60 ft, Gable Roofs Roof Slopes 27°≤Θ≤ 45°

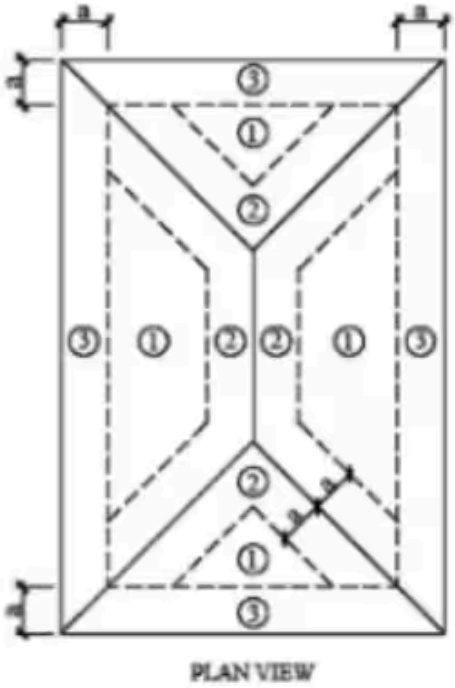


Hip roofs have one roof zone layout plan for all roof slopes, however, External pressure Coefficients graphs were updated for each.

ASCE 7-16



ASCE 7-22

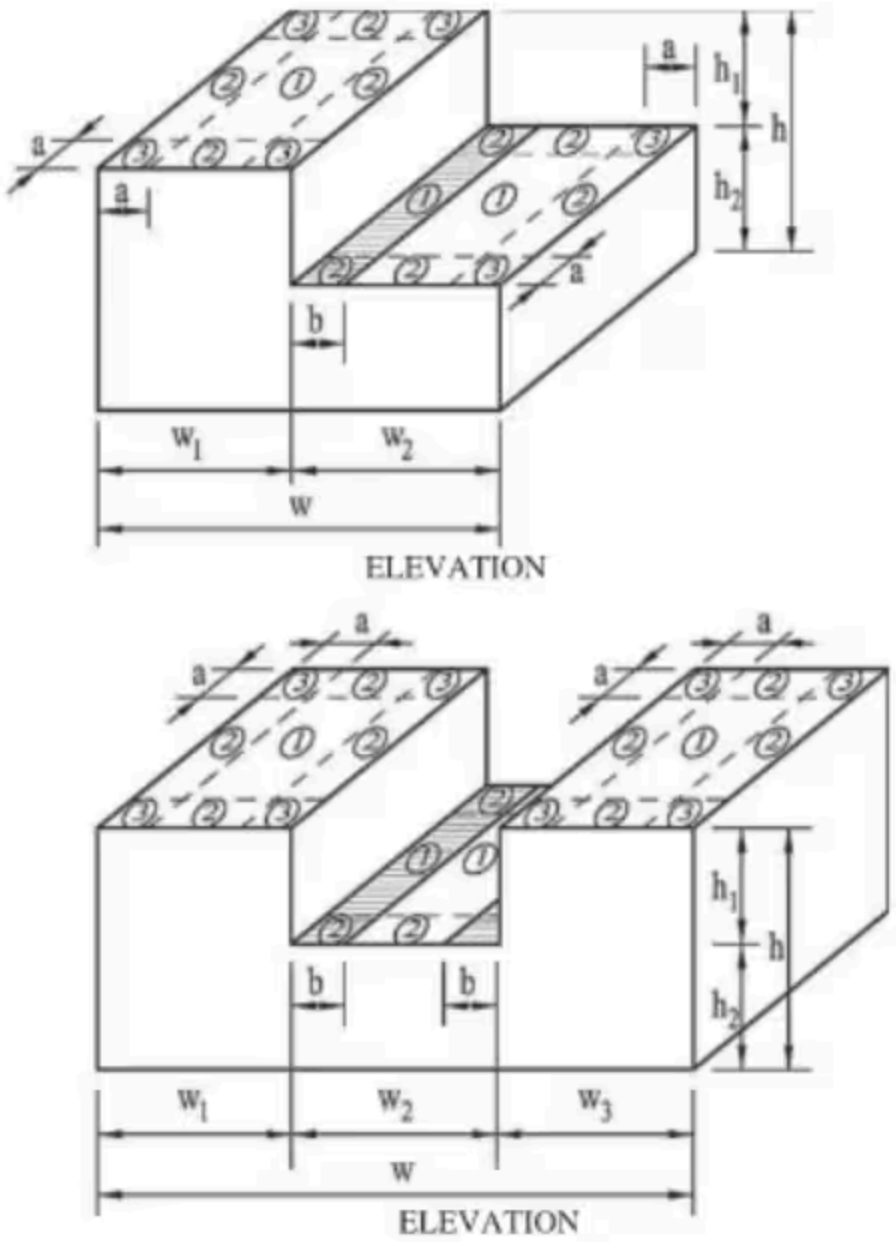


What is the Update? Roof Zones were revised for Stepped Roofs

Stepped roofs are where buildings have multiple flat roof levels, which are often seen on large hospitals, offices, and school buildings. While stepped roofs have wind uplift pressures and corresponding roof zones, it is important to note that wind on the lower roof is affected by the neighboring higher roof sections. At the intersection of the higher roof section with the lower roof section, the wind uplift pressures are lower.

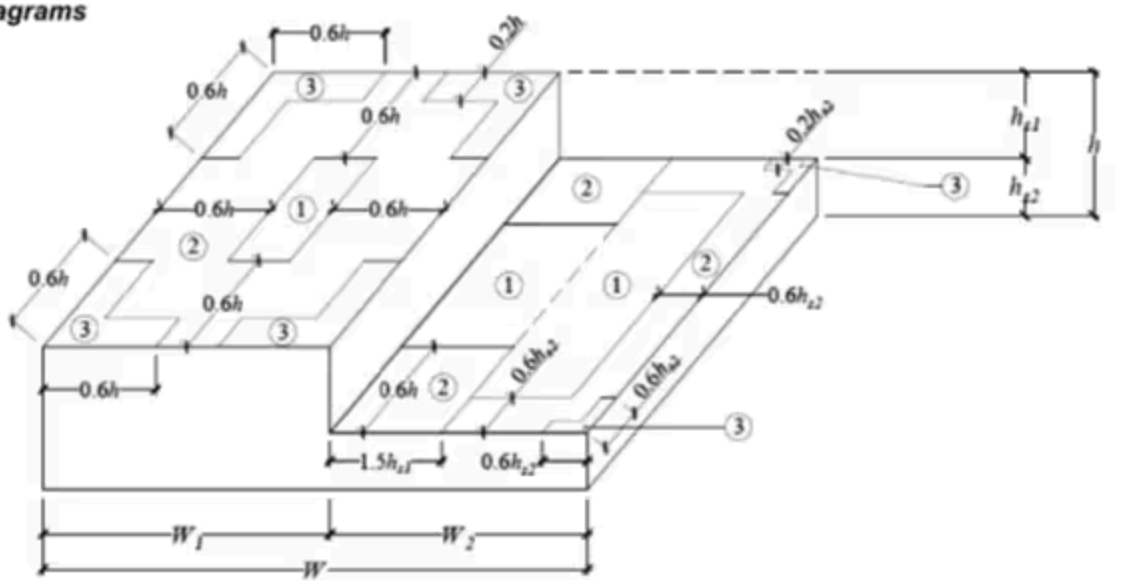
New diagrams have been inserted into Chapter 30 of the ASCE 7-22 version. The primary change to note is that the corner zones, zone 3, have been changed from square shapes to L shapes. This is a reflection of the standard roof zones which have been updated from square corners to L shaped corners in more recent versions of ASCE as well.

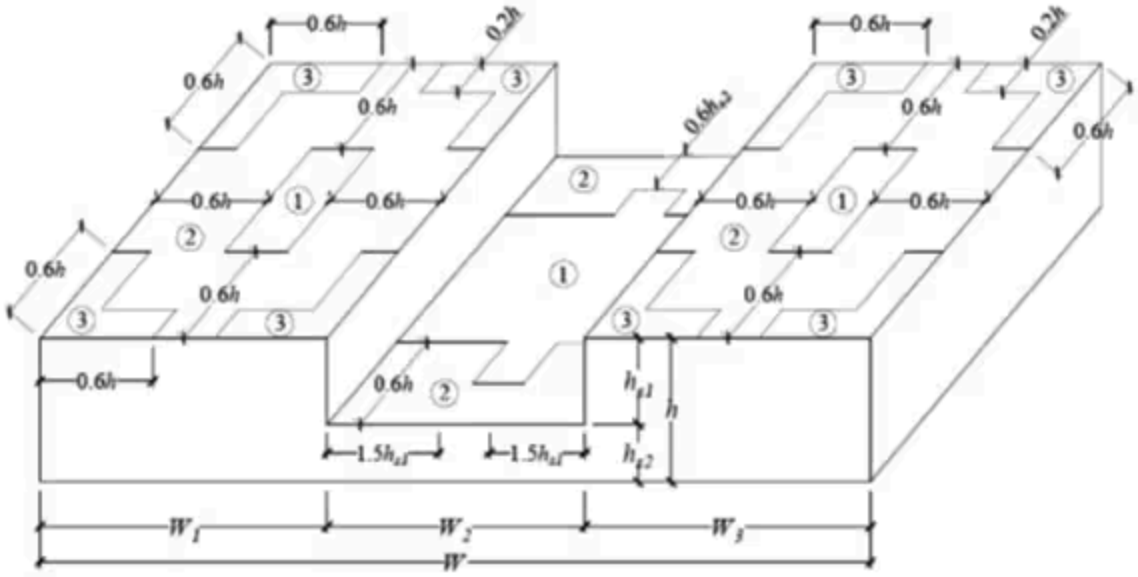
ASCE 7-16, Figure 30.3-3 Components and Cladding, h ≤ 60 ft



ASCE 7-22, Figure 30.3-3 Components and Cladding, h ≤ 60 ft

Diagrams





Notation

h = Mean roof height, ft (m)
W = Building width
Θ = Angle of the plane of the roof from horizontal, degrees